PROJECT AND TEAM INFORMATION

Project Title

Driver Monitoring System : Drowsiness & Mobile Distraction Detection with Mobile Integration

Student / Team Information

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Problem Statement

Driver drowsiness and mobile phone distraction are two issues that can affect road safety. Long driving hours and insufficient sleep may cause a driver to feel tired, lose concentration, close their eyes frequently, or yawn. Using a mobile phone while driving can also take attention away from the road. These behaviours may increase the risk of accidents, especially when the driver does not realize that their attention has decreased.

One challenge in driver monitoring is identifying these conditions through a camera. Lighting changes, facial movements, camera position, and obstruction of facial features can affect detection. A single blink or yawn does not always mean that the driver is drowsy. Similarly, detecting a mobile phone inside a vehicle does not necessarily mean that the driver is using it.

Therefore, there is a need for a Driver Monitoring System that can analyse facial features, use Machine Learning to classify the driver's condition, and detect possible mobile phone distraction. The system should combine these results to identify potentially risky behaviour and provide suitable alerts. Its performance must also be tested under different conditions to understand how accurately it can identify drowsiness.

Why We Built This Project

We built this project to explore how Computer Vision and Machine Learning can help identify unsafe driving behaviour through a camera. Our aim is to develop a practical, affordable system that can monitor driver drowsiness and mobile phone distraction without depending on expensive specialized hardware.

The system uses OpenCV for video processing and MediaPipe Face Mesh to identify facial landmarks. These landmarks are used to calculate the Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR), which help analyse eye closure and mouth opening, including possible yawning.

For driver-state classification, we prepared and augmented the dataset and trained several Machine Learning algorithms. These included K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Logistic Regression, Random Forest, Decision Tree, Naive Bayes, and Extra Trees. After comparing their performance, KNN was selected based on its validation F1-score.

The model comparison showed that KNN and SVM both achieved 85.44% accuracy, but KNN obtained a higher F1-score of 81.93%, compared with 81.01% for SVM. The final KNN test results showed approximately 86% accuracy on 71 test samples.

The project is currently around 50% complete. The dataset preparation, model training and comparison, and initial drowsiness-detection pipeline have been developed. The remaining work includes mobile phone detection, risk assessment, alerts, dashboard integration, and further testing.

Connecting AI with Human Decision-Making

Machine Learning can analyse data and identify patterns, but its predictions should not always be treated as final decisions. For example, a driver may blink normally or yawn without being dangerously drowsy. Similarly, a mobile phone may be visible inside a car without being used by the driver.

Our system is intended to combine the results of drowsiness detection and mobile phone detection. These results will be passed to a risk-assessment component to determine whether a potentially unsafe condition has been detected.

The project also plans to use a Streamlit dashboard to display the driver's status, detection results, and alerts. This will help users understand the output without needing to examine the Machine Learning model directly.

The purpose is to use AI as an assistance tool rather than a replacement for human judgment. Further testing will be needed to understand incorrect predictions and missed detections before the system can be considered suitable for practical use.

State of the Art / Current Solution

Existing driver-monitoring approaches use camera-based systems, wearable devices, vehicle sensors, and Advanced Driver Assistance Systems (ADAS). Camera-based techniques examine facial features such as eye opening, blinking, yawning, and head position to identify possible signs of drowsiness.

The Eye Aspect Ratio (EAR) is used to analyse eye opening, while the Mouth Aspect Ratio (MAR) can help identify mouth opening and possible yawning. Machine Learning algorithms can use extracted features or prepared datasets to classify the driver's state.

For mobile distraction detection, object-detection techniques can identify mobile phones visible in the camera feed. However, detection performance may depend on lighting conditions, camera angle, and how clearly the phone appears in the video.

Our project combines facial feature analysis with Machine Learning-based classification. We compared multiple algorithms and selected KNN based on the validation results. The mobile phone detection component, risk assessment, alert mechanism, and complete dashboard are still under development.

Our Solution

We are developing a Driver Monitoring System that monitors the driver's condition through a camera and is designed to detect both drowsiness and mobile phone distraction.

1. Real-Time Video Processing

Python and OpenCV are used to capture and process video from the camera. The processed frames provide input for the monitoring system.

Status: Basic video-processing pipeline developed.

2. Facial Landmark Detection

MediaPipe Face Mesh is used to detect important points on the driver's face. These landmarks provide the information required to analyse eye and mouth movements.

Status: Initial facial landmark detection developed.

3. EAR and MAR Feature Extraction

The system calculates two facial measurements:

- **Eye Aspect Ratio (EAR):** Helps analyse eye opening and prolonged eye closure.
- **Mouth Aspect Ratio (MAR):** Helps analyse mouth opening and possible yawning.

These features support the identification of possible drowsiness-related behaviour.

Status: Initial feature calculation developed.

4. Dataset Augmentation and Machine Learning

We prepared and augmented the drowsiness dataset before training and comparing several Machine Learning algorithms. The evaluated models included KNN, SVM, Logistic Regression, Random Forest, Decision Tree, Naive Bayes, and Extra Trees.

The model-comparison results are shown below.

```
Training: Random Forest
Training: SVM
Training: Logistic Regression
Training: Decision Tree
Training: KNN
Training: Naive Bayes
Training: Extra Trees
```

	Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
0	KNN	85.44	87.18	77.27	81.93
1	SVM	85.44	91.43	72.73	81.01
2	Logistic Regression	82.52	80.95	77.27	79.07
3	Random Forest	83.50	93.55	65.91	77.33
4	Extra Trees	80.58	87.50	63.64	73.68
5	Naive Bayes	78.64	80.56	65.91	72.50
6	Decision Tree	70.87	62.50	79.55	70.00

```
Selected using validation F1-score: KNN
```

KNN and SVM achieved the same accuracy in this comparison. However, KNN had the highest F1-score among the evaluated models and was selected for further testing.

Status: Dataset augmentation, model training, comparison, and KNN selection completed.

5. Final KNN Test Results

The selected KNN model was evaluated on a separate test set containing 71 samples.

Metric	Awake	Drowsy
Precision	0.84	0.89
Recall	0.93	0.77
F1-score	0.88	0.83
Test samples	40	31

The final test accuracy was approximately **86%**, with a weighted F1-score of approximately **86%**.

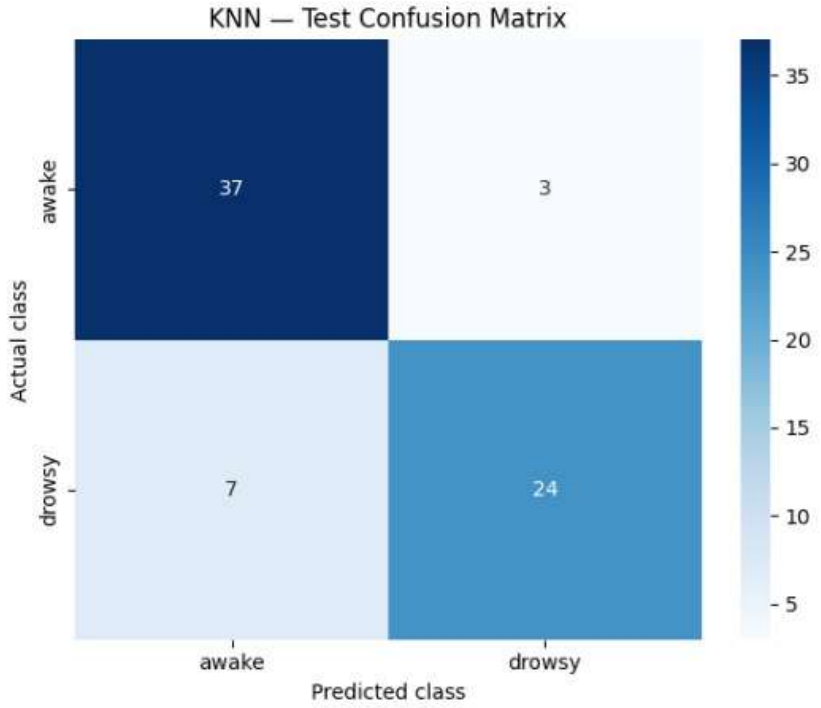
The confusion matrix showed that 37 awake samples and 24 drowsy samples were classified correctly. Three awake samples were incorrectly classified as drowsy, while seven drowsy samples were incorrectly classified as awake.

```
Selected model: KNN

Final test results:
              precision    recall  f1-score   support

       awake       0.84      0.93      0.88        40
      drowsy       0.89      0.77      0.83        31

    accuracy                           0.86        71
   macro avg       0.86      0.85      0.85        71
weighted avg       0.86      0.86      0.86        71
```



These results show that the model identifies many of the tested cases correctly, but some drowsy cases are still missed. Further evaluation and improvement are therefore required.

6. Mobile Distraction Detection

A separate object-detection module is being developed to identify mobile phones in the camera view. Its output will be integrated with the drowsiness-detection result.

Status: In progress.

7. Risk Assessment and Alerts

After the monitoring modules are integrated, their outputs will be combined to identify potentially risky driving conditions. The system is intended to provide audio or visual alerts when predefined risk conditions are detected. Optional mobile notifications may also be explored.

Status: Remaining development and integration work.

8. Current Progress

Approximately 50% of the project has been completed. The dataset has been prepared and augmented, multiple Machine Learning models have been trained and compared, and KNN has been selected based on its validation F1-score. The final KNN test evaluation has also been performed.

The remaining work includes completing mobile phone detection, integrating the monitoring modules, implementing risk assessment and alerts, finishing the Streamlit dashboard, and conducting further system testing.

Project Goals and Milestones

Project Goals

The main objective of this project is to develop a Driver Monitoring System that uses Computer Vision and Machine Learning to identify possible signs of drowsiness and mobile phone distraction.

The project aims to analyse facial landmarks, calculate EAR and MAR, and use a suitable classification algorithm to determine whether the driver is awake or showing signs of drowsiness. For this purpose, the dataset was augmented and several Machine Learning models were compared. KNN was selected after comparing the evaluation results.

Another objective is to develop mobile phone detection and combine its output with drowsiness classification. The integrated system will assess potential risks and provide alerts when required.

Finally, we aim to develop a Streamlit dashboard to display the monitoring results. The complete system will need to be tested under different lighting conditions, camera angles, and driver behaviours to evaluate its performance and limitations.

Project Milestones

Phase 1: System Planning and Technology Selection

We identified the driver-monitoring problem and selected the required tools, including Python, OpenCV, MediaPipe Face Mesh, Scikit-learn, NumPy, Pandas, and Streamlit.

Status: Basic planning and technology selection completed.

Phase 2: Dataset Preparation and Augmentation

We prepared the drowsiness dataset and applied data augmentation to increase the diversity of the training data. This dataset was then used for Machine Learning development.

Status: Dataset preparation and augmentation completed.

Phase 3: Model Training and Selection

We trained and compared seven Machine Learning algorithms: KNN, SVM, Logistic Regression, Random Forest, Decision Tree, Naive Bayes, and Extra Trees.

KNN and SVM both achieved 85.44% accuracy in the model comparison. KNN achieved a higher F1-score of 81.93%, so it was selected for further testing.

Status: Initial model comparison and KNN selection completed.

Phase 4: Final Model Evaluation and Drowsiness Detection

We evaluated the selected KNN model on 71 test samples and obtained approximately 86% accuracy. The camera-processing and facial feature extraction components provide the foundation for real-time drowsiness monitoring.

Further tests and improvements are required, particularly to reduce the number of drowsy cases classified as awake.

Status: Initial model evaluation and core drowsiness pipeline developed.

Phase 5: Mobile Phone Detection

The next stage is to develop the object-detection component for identifying mobile phones within the camera view. Its results will be combined with the drowsiness-detection output.

Status: In progress.

Phase 6: Risk Assessment and Alert Generation

The outputs from both detection modules will be combined to identify potential risks. Appropriate audio or visual alerts will then be implemented.

Status: Remaining integration and development work.

Phase 7: Streamlit Dashboard and Final Testing

The Streamlit dashboard will display the camera feed, driver status, detection results, and alerts. Once the modules are integrated, the complete system will be tested under different conditions.

The evaluation will examine correct classifications, missed drowsiness events, false alerts, and the limitations caused by lighting, facial movements, and phone visibility.

Status: Dashboard completion and full-system testing remain**.**

Expected Outcome

The expected outcome is a working proof-of-concept Driver Monitoring System that can analyse a camera feed, identify possible signs of drowsiness, and support mobile phone distraction detection.

The drowsiness module uses facial landmarks and EAR/MAR measurements. Multiple Machine Learning algorithms were trained and compared using the prepared and augmented dataset, and KNN was selected for further evaluation. The selected model achieved approximately 86% accuracy on the final test set containing 71 samples.

The completed system is expected to integrate drowsiness and mobile phone detection, combine their results through risk assessment, and provide suitable alerts. A Streamlit dashboard is also planned to make the monitoring results easier to view.

The current outcome includes dataset augmentation, model training and comparison, KNN selection, initial test evaluation, and development of the core drowsiness-detection pipeline. Mobile phone detection, complete module integration, alerts, dashboard completion, and final validation remain part of the work.

The project aims to demonstrate how Computer Vision and Machine Learning can be used together to identify potentially unsafe driving behaviour using accessible hardware and software. Further testing will be required to establish the system's reliability, and the prototype is not intended to replace certified vehicle safety systems.

References

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